

Relationships between lotic macroinvertebrate traits and responses to extreme drought

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SUMMARY

1. The prospect of increasing drought intensity in many river basins under climate change threatens the persistence of vulnerable freshwater species. Understanding how the traits of each species affect its resistance and resilience to drought may help to identify those species at most risk and elucidate the mechanisms by which impacts occur.
2. I analysed macroinvertebrate monitoring data collected from rivers across Australia's Murray–Darling Basin ($>10^6$ km²) during the middle and later stages of the recent decade-long Millennium Drought and the initial post-drought period. I tested the ability of eight traits, expressing aspects of life history, diet and environmental tolerance, to explain changes in the broad-scale prevalence (the proportion of sites with observed presence) of macroinvertebrate families during and after the drought.
3. The rate of basin-scale change in the riverine macroinvertebrate assemblage was least in the final stages of the drought. Immediately after the drought, the assemblage did not shift back towards its mid-drought state but instead moved further away. Eleven families that had a statistically significant fall in prevalence during the drought did not increase afterwards.
4. Negative responses to drought were associated with slower maturation, absence of atmospheric respiration, high rheophily and low thermophily. Positive responses to cessation of drought were associated with having a holometabolous life cycle, greater requirements for dissolved oxygen, high rheophily and low thermophily.
5. Because several traits were related to drought vulnerability, management to mitigate the adverse ecological effects of future droughts should consider a number of mechanisms by which drought has an effect. These include a loss of flowing water that supports rheophilous species, inadequate duration of wetting for species with a long aquatic phase, and effects of high temperature and hypoxia on species requiring cool conditions and well-aerated water.
6. A revival of research on the life histories of freshwater invertebrate species and more information on oxygen requirements and temperature and desiccation tolerance are needed to improve our ability to predict the effects of drought.

Keywords: climate change, drought, macroinvertebrate, river, trait

Introduction

Ecological responses to climate change are driven not only by shifts in average climatic conditions but also by alterations in the frequency and intensity of extreme events (Easterling *et al.*, 2000; Jentsch & Beierkuhnlein,

2008). These events include droughts, which have profound and complex consequences for aquatic ecosystems (Lake, 2011). Climatic modelling projects increasing drought during the 21st century over much of North and South America, southern Europe, the Middle East, southern Africa and Australia, as well as parts of east

Asia (Dai, 2013). Drying trends are already evident in some of these regions (Damberg & AghaKouchak, 2014; Vicente-Serrano *et al.*, 2014).

Australia's recent 'Millennium Drought' or 'Big Dry' illustrates the consequences of extreme drought for freshwater and floodplain environments. The Millennium Drought involved a series of rainfall deficiencies from the late 1990s to 2009, unprecedented in the historical record (Gergis *et al.*, 2012; Van Dijk *et al.*, 2013), and appears to have been primarily a product of multi-decadal climatic variability with additional contributions from anthropogenic climate change (Cai *et al.*, 2014). The drought ended in 2010 when a strong La Niña event brought heavy rainfall and widespread spring flooding (Hendon *et al.*, 2014). Its impacts on freshwater and associated ecosystems included marked declines in run-off and streamflow (Leblanc *et al.*, 2012) and broad-scale dieback of floodplain forests (MacNally *et al.*, 2011). The extensive lakes at the terminus of Australia's largest river system, the Murray–Darling, contracted to below sea level and became salinised and acidified (Kingsford *et al.*, 2011). Drying resulted in local fish extinctions (Bond & Lake, 2005), and emergency interventions were needed to save a number of populations of threatened fish species (Hammer *et al.*, 2013). Despite a return of cooler and wetter conditions in 2010, recovery of fish and frog populations remains incomplete (MacNally *et al.*, 2014; Wedderburn, Barnes & Hillyard, 2014).

Freshwater invertebrates were among the organisms affected by the Millennium Drought, with many taxa that are adapted to cooler environments and flowing water suffering broad-scale declines across south-eastern Australia between the 1990s and 2010 (Chessman, 2009, 2012; Thomson *et al.*, 2012). The extent to which these taxa have rebounded since the drought is not yet clear, but regardless of the current level of recovery, those with low drought resistance (ability to withstand drought) or resilience (ability to recover from drought) are likely to be vulnerable to the expected future trend of increasing drought frequency and severity.

The resistance and resilience of freshwater invertebrates to drought have been associated with numerous characteristics such as the ability to tolerate poor water quality, short duration of life-history stages that require surface water, possession of life-cycle stages that tolerate drying, and strong powers of dispersal (Iversen *et al.*, 1978; Boulton, 2003; Griswold *et al.*, 2008). However, little assessment has been made of how particular traits relate to the responses of freshwater invertebrates to extreme drought, especially over large spatial extents (Boulton & Lake, 2008). Such information should be

useful not only for identifying species at risk from future drought, but also for providing insight into the processes through which vulnerable species are affected, and thereby informing management responses. These topics have global relevance given projections of increasing drought intensity in many of the world's major river basins (Prudhomme *et al.*, 2014; Van Huijgevoort *et al.*, 2014).

Here, I tested the ability of eight traits to explain changes in the broad-scale prevalence of riverine macroinvertebrate taxa both during and after the Millennium Drought, extending previous research that considered invertebrate responses to climatic warming and drying in relation to the traits of rheophily and thermophily (Chessman, 2009, 2012). For the present work, I analysed data from a 'river health' monitoring programme that sampled running-water macroinvertebrates throughout the Murray–Darling Basin ($>10^6$ km²) in south-eastern Australia from 2004 to 2012, spanning the drought's middle and final phases, when impacts on river hydrology became most intense, plus the immediate post-drought period. My aims were (i) to assess the extent to which macroinvertebrate taxa declined during the drought period and subsequently recovered and (ii) to explore how the way each taxon fared, both during the drought and afterwards, related to its traits. I selected a range of traits, expressing aspects of life history, diet and environmental tolerances, whose values have been reported or could be estimated for most or all of the taxa included in the analysis. I then hypothesised how each trait would relate to the drought and post-drought responses of macroinvertebrate taxa (Table 1).

Methods

Study area

The Murray–Darling Basin covers 1 061 469 km² between the latitudes of 24 and 38°S and includes the Australian Capital Territory and portions of four States (New South Wales, Queensland, South Australia and Victoria). It ranges in altitude from sea level to 2228 m and contains numerous wetlands, lakes and streams, including Australia's three longest rivers – the Darling (2740 km), Murray (2530 km) and Murrumbidgee (1690 km). Most of the basin is arid or semi-arid and therefore most rivers outside of the uplands cease to flow or dry completely during extended periods of low precipitation, unless sustained by releases from dams. The basin also has a modest annual average discharge considering its area: 4700 GL, which would be 12200 GL

Table 1 Macroinvertebrate traits included in the study and their hypothesised relationships to how taxa fare during and after drought

Trait name	Measure	Hypothesis	Rationale
Maturation time	Age at maturity (months)	Fast-maturing taxa fare better during drought and slow-maturing taxa fare better after drought	Water is more intermittent during drought and small-bodied taxa with faster life cycles are better adapted to more temporary waters (Barclay, 1966; Williams, 1996; Arscott <i>et al.</i> , 2010)
Holometaboly	Occurrence of a pupal stage in the life cycle	Holometabolous taxa fare worse during drought and better after drought	As above
Body size	Maximum length (mm)	Larger-bodied taxa fare worse during drought and better after drought	As above
Dietary generalisation	Number of major food categories in the diet	Dietary generalists fare better during drought and worse after drought	Taxa with generalised diets are better adapted to more temporary waters (Williams, 1996)
Atmospheric respiration	Occurrence of anatomical and behavioural adaptations for accessing atmospheric oxygen in aquatic life-history stages	Air-breathing taxa fare better during drought and non-air-breathing taxa fare better after drought	Dissolved oxygen levels are often reduced during drought (Boulton & Lake, 1990; Magoulick & Kobza, 2003; Acuña <i>et al.</i> , 2005)
Dissolved oxygen (DO) requirement	Minimum DO concentration of occurrence (mg L ⁻¹)	Taxa that tolerate lower DO concentrations fare better during drought and less tolerant taxa fare better after drought	As above
Rheophily	Association with fast-flowing water	Rheophiles fare worse during drought and better after drought	Stream flows are reduced or cease during drought (Lake, 2003; Bond, Lake & Arthington, 2008)
Thermophily	Association with higher temperatures	Thermophiles fare better during drought and worse after drought	Drought is often associated with higher water temperatures (Lake, 2003; Magoulick & Kobza, 2003)

without water diversions (CSIRO, 2008). It is an important agricultural region, containing about 65% by area of Australia's irrigated pastures and crops and 39% by value of the nation's farm production (ABS, 2008).

Annual mean air temperature across the entire basin has been continuously above the Australian Bureau of Meteorology's baseline (1961–90 mean) since 1997, although only slightly so during 2010–12 (Fig. 1). Annual rainfall has been below the baseline since 2001, except in 2010–12. As a consequence of higher temperature and reduced rainfall, stream flows decreased severely over the course of the Millennium Drought, with many streams reducing to stagnant pools or drying altogether (B.C. Chessman, personal observations).

Macroinvertebrate monitoring data

I assessed changes in macroinvertebrate assemblages during and after the Millennium Drought with data from the Sustainable Rivers Audit (SRA: Davies *et al.*, 2010), a monitoring programme that sampled macroinvertebrates from rivers (but not reservoirs, natural lakes or wetlands) across the entire Murray–Darling Basin from late 2004 until mid-2012. Individual catchments within the basin were visited on a rotating basis, with each sampled once every 2 years in either autumn–early

winter or spring–early summer. Thus, there were four sampling rounds comprising the periods 2004–06, 2006–08, 2008–10 and 2010–12, with the third round ending as the drought was breaking and the last round falling in the immediate post-drought period. Macroinvertebrate sampling sites within each catchment were selected by a stratified random process wherein the catchment was divided into altitudinal zones and sites were chosen randomly from a defined stream network within each zone, but with restrictions on the proximity of sites to one another. Backup sites were selected to replace sites that were dry when visited. Altogether, 779–816 sites were sampled across the basin in each round. Some 7% of sites were sampled in all four rounds but 89% were sampled once only.

Two macroinvertebrate samples were collected from each site on each sampling occasion. For sites with fast-flowing water, one sample was collected from riffle habitat by kick sampling and one from slow-flowing or still water near the stream edge by sweep sampling, in both cases with a hand net (250-µm mesh) over an estimated distance of 10 m. If flowing riffles were absent, two edge-water samples were collected. Methods varied geographically because individual State agencies followed different protocols; for example, in South Australia, samples were preserved and subsampled randomly in the

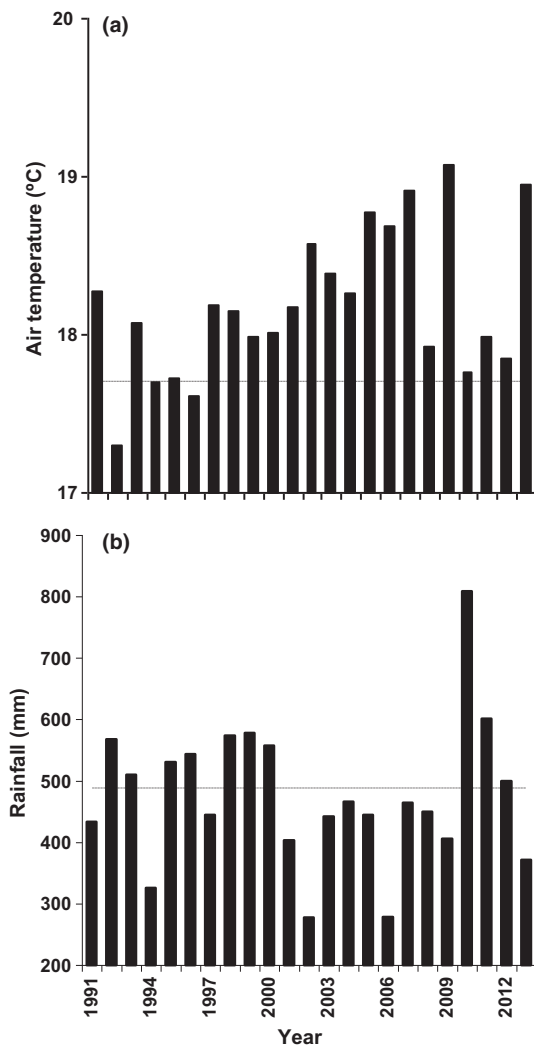


Fig. 1 (a) mean air temperature and (b) total rainfall for the Murray–Darling Basin in each year from 1991 to 2013, as reported by the Australian Bureau of Meteorology (www.bom.gov.au; accessed May 2014). Horizontal lines indicate baseline averages (1961–90).

laboratory, whereas in other States, live samples were subsampled selectively on site.

Data provided by the Murray–Darling Basin Authority from its SRA database comprised the number of specimens of each macroinvertebrate taxon recorded from each sample. These taxa ranged from genus to phylum, although most were families, and taxonomy was sometimes outdated and inconsistent across the basin. For uniformity in analysis, I ignored all specimens not identified at least to family, updated names to currently recognised families where applicable, and grouped families that had sometimes been identified with the present, narrow family definitions and sometimes with old, broader definitions. I also omitted extremely rare

families recorded in fewer than 10 combinations of site and time, leaving 125 families or family groups for analysis. All name changes and groupings are listed in Table 2. For simplicity, all taxa used in the analysis are referred to as families hereafter.

Assignment of macroinvertebrate traits

In most cases, trait values were assigned with information from the literature and websites, with emphasis on Australian information where possible, especially Chessman (1986), Gooderham & Tsyrlin (2002), Schäfer *et al.* (2011) and *Identification and Ecology of Australian Freshwater Invertebrates* (www.mdfr.org.au/bugguide/index.htm, accessed October 2013). Where information varied among sources or for different taxa within a family, I used average values.

Literature and online sources were consulted for traits related to life history (age at maturity, the presence or absence of a holometabolous life cycle and the maximum body length attained), feeding and respiration. I summarised dietary generalisation as the number out of five major food categories (animal material, fine organic deposits, periphyton, plankton plus seston and plant material) in the diet of each family. Use of atmospheric respiration was characterised as the percentage of all respiratory modes of aquatic life-cycle stages of each family that involved access to atmospheric oxygen (possible modes being aerial, branchial, cutaneous, plastron and pulmonary). An estimate of minimum dissolved oxygen (DO) concentration tolerated by aquatic stages of each family was obtained from calculations made by Chessman & McEvoy (2012).

The preference of each family for warmer conditions (thermophily) and rapid flow (rheophily) was estimated by considering its occurrence in 8928 biomonitoring samples collected in the Australian Capital Territory and New South Wales between 1994 and 2010 (Chessman, 2012). Thermophily was estimated as the mean water temperature associated with all samples in which a family was collected for which contemporaneous temperature data were available, divided by the mean temperature for all samples with temperature data. Because data on current velocity were not available for any of the samples, rheophily was estimated from the habitats from which samples were taken. Habitats were scored in order of likely increasing velocity as 0 for pool edges and other still waters, 1 for glides, 2 for runs, 3 for riffles, 4 for rapids and 5 for cascades and waterfalls. The mean score of samples in which each family was detected was divided by the mean for all scored samples.

Table 2 Macroinvertebrate families and family groups used in the analysis, with taxon names in the SRA database that were treated as belonging to each family or group. Statistically significant ($P < 0.05$) changes in prevalence during the Millennium Drought (between 2004–06 and 2008–10), after the drought (between 2008–10 and 2010–12), and over the whole monitoring period (between 2004–06 and 2010–12) are indicated as F (significant fall) or R (significant rise)

Family or group	Database taxon or taxa	Drought change	Post-drought change	Overall change
Aeshnidae group	Aeshnidae, Telephlebiidae	F		F
Ameletopsidae	Ameletopsidae			
Argiolestidae	Megapodagrionidae	F		F
Athericidae	Athericidae			
Atriplectididae	Atriplectididae			
Atyidae	Atyidae	F		F
Austroperlidae	Austroperlidae	F	R	
Baetidae	Baetidae	F		
Belostomatidae	Belostomatidae			R
Blephariceridae	Blephariceridae			R
Caenidae	Caenidae		F	F
Calamoceratidae	Calamoceratidae			
Calocidae	Calocidae	F	R	
Carabidae	Carabidae			
Ceratopogonidae	Ceratopogonidae	R	F	F
Chaoboridae	Chaoboridae			F
Chiltoniidae	Ceinidae			
Chironomidae	Aphroteniinae, Chironomidae, Chironominae, Chironomini, Diamesinae, Orthocladiinae, Podonominae, Pseudochironomini, Tanyptodinae, Tanytarsini,	R	F	
Chrysomelidae	Chrysomelidae		F	F
Coenagrionidae	Coenagrionidae			
Coloburiscidae	Coloburiscidae	F	R	
Conoesucidae	Conoesucidae	F	R	
Corallanidae	Corallanidae		F	F
Corbiculidae	Corbiculidae			F
Corduliidae group	Cordulephyidae, Corduliidae, Hemicorduliidae, Synthemistidae			F
Cordylorphoridae	Clavidae			
Corixidae	Corixidae			
Corydalidae	Corydalidae			
Crambidae	Pyrallidae		F	F
Culicidae	Culicidae			
Curculionidae	Curculionidae			
Diphlebiidae	Diphlebiidae			
Dixidae	Dixidae			
Dolichopodidae	Dolichopodidae		R	R
Dugesidae	Dugesidae		F	F
Dytiscidae	Dytiscidae	R	R	R
Ecnomidae	Ecnomidae	R		
Elmidae	Elmidae	F		
Empididae	Empididae	F	R	
Ephydriidae	Ephydriidae	R		R
Eustheniidae	Eustheniidae			
Gelastocoridae	Gelastocoridae			
Gerridae	Gerridae			
Glossiphoniidae	Glossiphoniidae			F
Glossosomatidae	Glossosomatidae			
Gomphidae group	Gomphidae, Lindaniidae [sic]		F	F
Gordiidae	Gordiidae			
Gripopterygidae	Gripopterygidae			
Gyrinidae	Gyrinidae			
Halipilidae	Halipilidae			
Hebridae	Hebridae		R	R

Table 2 (Continued)

Family or group	Database taxon or taxa	Drought change	Post-drought change	Overall change
Helicophidae	Helicophidae			
Helicopsychidae	Helicopsychidae			
Heteroceridae	Heteroceridae	R		R
Hirudinidae	Richardsonianidae			
Hydraenidae	Hydraenidae	R	R	R
Hydridae	Hydridae			
Hydrobiosidae	Hydrobiosidae		R	
Hydrometridae	Hydrometridae	R	R	R
Hydrophilidae	Hydrochidae, Hydrophilidae	R	R	R
Hydropsychidae	Hydropsychidae		R	
Hydroptilidae	Hydroptilidae	F	F	F
Hygrobiidae	Hygrobiidae	F		F
Hyriidae	Hyriidae			
Isostictidae	Isostictidae			
Janiridae	Janiridae			
Leptoceridae	Leptoceridae	F	R	
Leptophlebiidae	Leptophlebiidae			
Lestidae	Lestidae			
Libellulidae group	Libellulidae, Urothemistidae		F	F
Limnephilidae	Limnephilidae		F	
Lymnaeidae	Lymnaeidae		F	
Mesoveliidae	Mesoveliidae			F
Muscidae	Muscidae			
Nannochoristidae	Nannochoristidae		R	R
Nanophyidae	Brentidae		F	
Naucoridae	Naucoridae	F		F
Neoniphargidae	Neoniphargidae			
Nepidae	Nepidae	F	R	
Nevrothidae	Nevrothidae [sic]			
Noteridae	Noteridae	R		R
Notonectidae	Notonectidae			
Notonemouridae	Notonemouridae			
Ochteridae	Ochteridae			
Odontoceridae	Odontoceridae			
Oniscidae	Oniscidae			
Oniscigastridae	Oniscigastridae			
Osmylidae	Osmylidae		F	F
Palaemonidae	Macrobrachium, Palaemonidae		F	
Paramelitidae	Paramelitidae			
Parastacidae	Parastacidae			
Perthiidae	Perthiidae			
Philopotamidae	Philopotamidae			
Philorheithridae	Philorheithridae	F	R	
Phreatoicidae	Phreatoicidae			F
Physidae	Physidae		F	F
Planorbidae	Ancylidae, Planorbidae			F
Platycnemididae	Protoneuridae		F	F
Pleidae	Pleidae			
Polycentropodidae	Polycentropodidae			
Pontogeneiidae	Eusiridae			
Psephenidae	Psephenidae	F		
Psychodidae	Psychodidae		R	R
Ptilodactylidae	Ptilodactylidae			
Saldidae	Saldidae			R
Sciomyzidae	Sciomyzidae	R		
Scirtidae	Scirtidae			
Sialidae	Sialidae	F		F
Simuliidae	Simuliidae		R	R

Table 2 (Continued)

Family or group	Database taxon or taxa	Drought change	Post-drought change	Overall change
Siphonotidae	Siphonotidae			
Sisyridae	Sisyridae	F		
Sphaeriidae	Sphaeriidae		F	
Staphylinidae	Staphylinidae	R	F	
Stratiomyidae	Stratiomyidae			F
Synlestidae	Synlestidae			
Syrphidae	Syrphidae	R		
Tabanidae	Tabanidae			
Talitridae	Talitridae			
Tasimiidae	Tasimiidae		R	R
Tateidae	Hydrobiidae		F	F
Temnocephalidae	Temnocephalidae	R		R
Thaumaleidae	Thaumaleidae			
Thiaridae	Thiaridae			
Tipulidae	Tipulidae		R	R
Veliidae	Veliidae	R		R

Analysis of temporal changes

Because sampling and subsampling methods were spatially variable and at most semiquantitative, I ignored counts of numbers of individuals per family and expressed the occurrence of each family as the proportion of sites in which it was recorded in each sampling round (hereafter 'family prevalence'). I calculated the cumulative macroinvertebrate assemblage change between the first sampling round in 2004–06 and subsequent rounds by applying the Bray–Curtis dissimilarity coefficient to prevalence data for all families. I did this separately for untransformed prevalence and for prevalence transformed to square roots and fourth roots in order to test the effect of progressive down-weighting of the more common families. I used the large-sample χ^2 test for equality of proportions (Sheskin, 2007) to test for statistically significant ($P < 0.05$) changes in the prevalence of individual families, both during the drought (i.e. between 2004–06 and 2008–10) and immediately afterwards (between 2008–10 and 2010–12). I related these changes to individual traits by calculating Spearman rank correlation coefficients (r_s), with the change in prevalence of each family expressed in both absolute terms ($\Delta P_a = P_f - P_i$, where P_f is final prevalence and P_i is initial prevalence) and relative terms ($\Delta P_r = (P_f - P_i)/P_i$). I did not apply a Bonferroni adjustment for multiple testing in any of these analyses because of its logical and practical problems (Perneger, 1998; Moran, 2003), but rather considered the implications of multiple testing when interpreting results. All statistical tests were performed with STATISTICA version 12 (www.statsoft.com).

Results

The basin-scale composition of the macroinvertebrate fauna diverged further from its 2004–06 composition with each successive sampling round (Fig. 2), the rate of change being slowest (i.e. the rise in Bray–Curtis dissimilarity being least) in the late-drought period, between 2006–08 and 2008–10. Transformation of prevalence data to square or fourth roots made little difference to these temporal patterns (Fig. 2).

During the drought period, between the first and third sampling rounds, some 12% of analysed macroinvertebrate families had a significant ($P < 0.05$) rise in prevalence (χ^2 test) and 15% had a significant fall (Fig. 3; Table 2). In the transition from drought to wet conditions, between the third and final sampling rounds, 17% of families rose significantly and 16% fell significantly. Overall, from the first to the final sampling round, 15% rose significantly and 22% fell significantly. In each case, the total percentage of families with significant changes (27–37%) was substantially above the 5% expected by chance when testing for significance at $\alpha = 0.05$. Changes in the prevalence of individual families during the drought were weakly but significantly correlated with post-drought changes when changes were expressed in absolute terms ($r_s = -0.24$; $P = 0.008$), but the correlation was not quite significant when changes were in relative terms ($r_s = -0.17$; $P = 0.054$) (see Fig. 4). Eleven families had significant post-drought changes in prevalence that were the opposite of those observed during the drought, whereas another 11 had a significant fall in prevalence during the drought and no significant post-drought increase (Table 2).

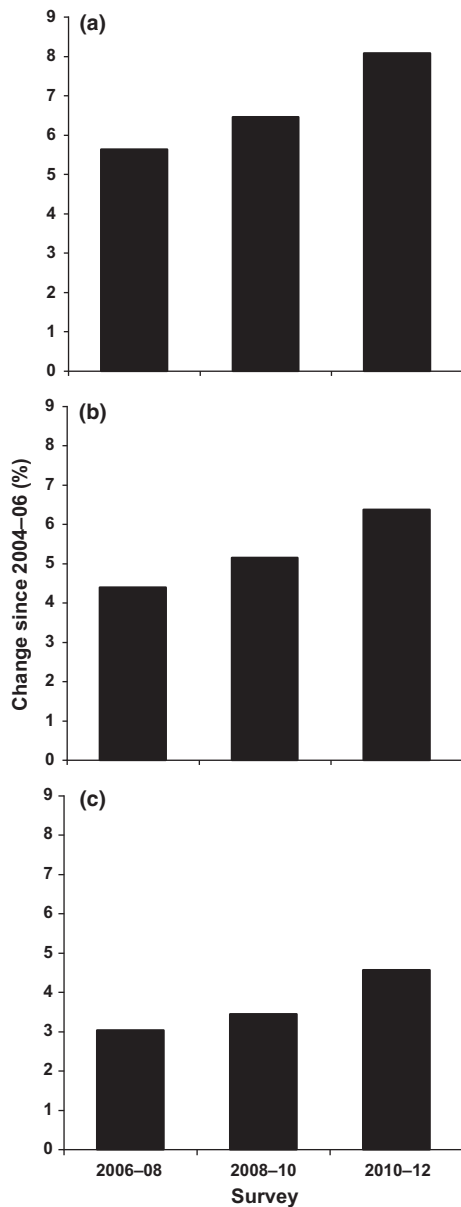


Fig. 2 Cumulative basin-scale change in the macroinvertebrate fauna between the first SRA survey in 2004–06 and subsequent surveys in 2006–08, 2008–10 and 2010–12. Change is quantified as the Bray–Curtis dissimilarity index applied to family prevalence data and expressed as a percentage: (a) untransformed data; (b) square-root-transformed data; (c) fourth-root-transformed data.

Intercorrelations of the traits considered were generally low. However, atmospheric respiration, DO requirement, rheophily and thermophily were all associated with one another, with absolute r_s values of between 0.30 and 0.58. Families with high rheophily tended to lack atmospheric respiration, have high DO requirements and favour cooler environments, while families with low rheophily tended to be capable of atmospheric

respiration, tolerate hypoxia and favour warmer environments.

Statistically significant correlations between traits and prevalence changes occurred in 12 of 32 instances (Table 3), over seven times as often as expected by chance, and always accorded with *a priori* hypotheses (Table 1). Prevalence changes during the drought were positively associated with atmospheric respiration and thermophily and negatively associated with maturation time and rheophily. Post-drought changes in prevalence were positively associated with a holometabolous life cycle, greater DO requirement and rheophily, and negatively associated with thermophily.

Discussion

Macroinvertebrate assemblage changes during and after drought

The Millennium Drought and the heavy rainfall that followed it produced dramatic hydrological fluctuations in the Murray–Darling River system, ranging from the lowest runoff on record to extensive flooding (Chiew *et al.*, 2014). These changes were almost certainly the primary cause of the broad-scale shifts in the macroinvertebrate fauna, revealed by both the analysis of assemblage composition and the significance tests for individual families. A decline in water diversions during the drought (Leblanc *et al.*, 2012) may also have had some influence, but this was driven chiefly by the drought and therefore cannot be separated from it. The pace of faunal change slowed towards the end of the drought, perhaps because its effects had largely run their course by that time, and accelerated again in the transition from the drought to the initial post-drought period. Although some families had significant post-drought changes in prevalence that reversed those observed during the drought, the overall macroinvertebrate assemblage continued to shift away from its initial (2004–06) composition through to the final sampling round in 2010–12.

In some cases, the composition of stream macroinvertebrate assemblages can change substantially during drought yet recover to resemble pre-drought composition within a year or less of the drought ending (Lake, 2003). However, this rapid recovery may not occur after extreme or prolonged drought, perhaps because of enduring habitat changes, local extinctions, constraints on recolonisation and drought-facilitated species invasions. For example, Bêche *et al.* (2009) found that stream macroinvertebrate assemblage composition had not

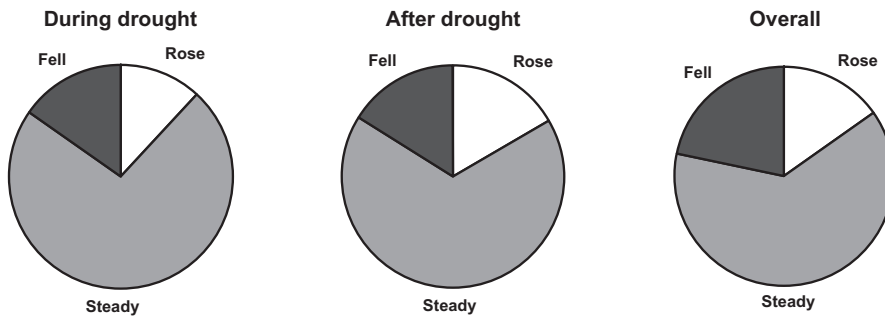


Fig. 3 Proportions of macroinvertebrate families that significantly rose or fell in prevalence (χ^2 test; $P < 0.05$) or remained steady ($P > 0.05$) during the Millennium Drought (between 2004–06 and 2008–10), after the drought (between 2008–10 and 2010–12), and overall (between 2004–06 and 2010–12).

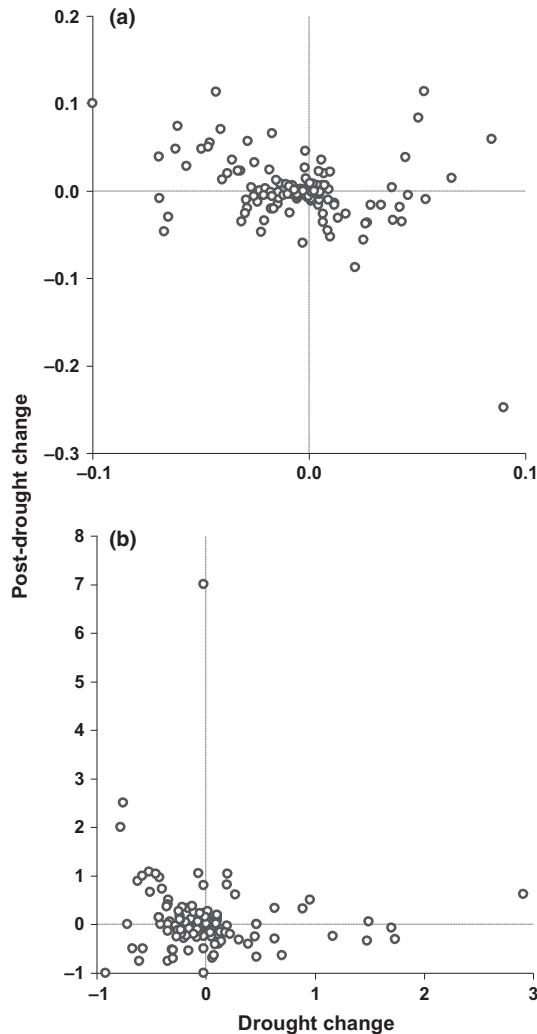


Fig. 4 Relationships between the change in prevalence of each macroinvertebrate family during the Millennium Drought (between 2004–06 and 2008–10) and its change after the drought (between 2008–10 and 2010–12): (a) absolute changes; (b) relative changes.

recovered to its pre-drought state 12 years after a 5-year drought in California, and Bogan & Lytle (2011) observed a compositional shift that endured for at least 4 years after the transition of a formerly perennial desert

stream in Arizona to intermittent flow during a 5-year drought. Persistent effects of historically extreme drought have also been observed for individual taxa: for instance, Resh (1992) found that the population age structure of a caddisfly species took 10 years to return to its prior status after severe drought in a Californian springbrook, and Sponseller *et al.* (2010) found that hydropsychid caddis flies did not recover during 9 years after a record drought in an Arizonan desert stream. Unfortunately, the SRA did not cover a full a hydro-meteorological cycle from the start of one multi-year drought to the onset of the next. Consequently, the accumulating basin-scale change in the macroinvertebrate assemblage observed from the mid-drought through to the immediate post-drought period could be part of a supra-decadal rotation whereby the assemblage will eventually return to its initial state, rather than representing an irreversible faunal shift. Monitoring of some sites is continuing under other programmes and will provide further information on longer-term trends.

One factor that may have inhibited post-drought recovery of macroinvertebrates in the Murray–Darling Basin was the occurrence of severe ‘blackwater’ events in the drought’s aftermath. The post-drought inundation of large quantities of leaf litter that had accumulated in riparian zones and on floodplains during the drought led to widespread water-column hypoxia as a result of microbial decomposition (Whitworth, Baldwin & Kerr, 2012). Blackwater conditions in 2010–11 affected populations of fish and decapod crustaceans (King, Tonkin & Lieshcke, 2012; McCarthy *et al.*, 2014) and possibly other macroinvertebrates.

Drought responses and traits

Several factors limited the identification of traits associated with drought resistance and resilience. Data constraints restricted the analysis to family level, but finer taxonomic resolution would have been desirable because

Table 3 Significant Spearman coefficients of rank correlation between macroinvertebrate traits and absolute and relative changes in family prevalence during the Millennium Drought (between 2004–06 and 2008–10) and afterwards (between 2008–10 and 2010–12). Probability values are in parentheses

Trait	Drought change (absolute)	Drought change (relative)	Post-drought change (absolute)	Post-drought change (relative)
Maturation time	–0.22 (0.014)			
Holometaboly			0.24 (0.007)	
Body size				
Dietary generalisation				
Atmospheric respiration		0.18 (0.041)		
DO requirement			0.23 (0.019)	0.27 (0.006)
Rheophily	–0.25 (0.005)	–0.26 (0.003)	0.28 (0.002)	0.32 (<0.001)
Thermophily		0.21 (0.017)	–0.34 (< 0.001)	–0.31 (<0.001)

trait values can vary substantially among species or genera within a family (Monaghan & Soares, 2013). Further, only changes in prevalence (the proportion of sites with observed presence) could be assessed and not changes in abundance. In addition, the SRA did not sample before the Millennium Drought or during its early stages, and consequently, initial responses of the macroinvertebrate assemblage, which might have been substantial, could not be included in the analysis. Moreover, the last sampling round began very early in the post-drought period and, as a result, the final assemblage change probably did not reflect the full post-drought response. Finally, the compilation of trait data revealed a scarcity of information for Australian freshwater invertebrates, even at family level. Some traits that were potentially highly relevant, such as dispersal capacity and ability of different life-history stages to withstand drying, were omitted from the analysis because sufficient information could not be found to assign values to enough families with confidence. Notwithstanding these limitations, statistically significant associations with changes in the prevalence of macroinvertebrate taxa were obtained for two-thirds of the traits considered, and these always supported the study hypotheses.

Macroinvertebrates that require less time to reach maturity tended to fare better during the Millennium Drought (i.e. were more likely to increase in prevalence and less likely to decrease), while those with a pupal stage in the life cycle tended to do better after the drought. Small body size, rapid growth and short life spans are common attributes of inhabitants of temporary waters (Williams, 1996), and invertebrates with faster maturation may have recruited more successfully during the drought by being able to complete development from egg to adult during brief flow episodes or in temporary pools. However, maximum body length was not significantly associated with changes in prevalence

either during or after the Millennium Drought. For Californian streams, Bèche & Resh (2007) found that drier years were associated with higher relative abundance of invertebrates with very small maximum body size (<5 mm), moderate body size (10–20 mm) or very large body size (>40 mm), depending on the region, whereas wetter years were associated with a greater proportion of invertebrates with small body size (5–10 mm). Hence, relationships between body size and the fate of taxa during and after drought may be too complex and spatially variable to be evident at the scale of the whole Murray–Darling Basin. In addition, the reported maximum body size within a family may be a poor indicator of the actual size distribution of constituent individuals (Orlofske & Baird, 2014).

Although Williams (1996) proposed that generalist feeding is adaptive for life in temporary waters, dietary generalism was not related to changes in prevalence either during or after the drought. This suggests that specialised feeding is not a major factor in determining drought vulnerability. It is possible that dietary diversity was described too crudely by considering just five broad food categories, with only one category being assigned to 78% of the families considered. Nevertheless, the present results accord with those of Ledger *et al.* (2013), who found no evidence that simulated supra-seasonal drought in artificial streams favoured trophic generalists over specialists.

The association of changes in prevalence with capacity for atmospheric respiration and tolerance of low DO concentration suggests that scarcity of DO may have been an important mechanism of drought impact. Dissolved oxygen declined during the Millennium Drought (Rose, Metzeling & Catzikiris, 2008; Thomson *et al.*, 2012), probably because of high temperatures, decomposition of accumulated organic matter and limited re-aeration under stagnant conditions (cf. Boulton

& Lake, 1990; Acuña *et al.*, 2005). Invertebrates that tolerate low DO concentrations commonly respire atmospheric air (Justus *et al.*, 2014), and drought often leads to a dominance of such taxa (Boulton & Lake, 1992b; Acuña *et al.*, 2005; Bèche & Resh, 2007; Verkaik *et al.*, 2013).

Rheophilous families fared poorly during the drought and better afterwards. This is also a frequent observation in drought-affected streams (Boulton & Lake, 1992a; Acuña *et al.*, 2005; Rose *et al.*, 2008; Pace, Bonada & Prat, 2013). A requirement for current may relate to a range of associated resources, such as the supply of oxygen to respiratory surfaces (Golubkov, Tiunova & Kocharina, 1992) and provision of a continuous food supply to filter feeders (Wallace & Merritt, 1980). Filter-feeding larvae of Simuliidae and Hydropsychidae, which have previously been recorded to colonise rapidly or increase in abundance after drought (Harrison, 1966; Canton *et al.*, 1984; Boix *et al.*, 2010), had significant rises in prevalence across the Murray–Darling Basin in the initial post-drought period.

The positive relationship between thermophily and changes in prevalence during the drought, and negative relationship afterwards, accords with the interannual pattern of mean annual air temperatures (Fig. 1). Little information is available on thermal tolerances of freshwater invertebrates in the Murray–Darling Basin, but upper limits for those occurring at similar latitudes in south-western Australia can be as low as 20 °C (Stewart *et al.*, 2013). Since summer stream temperatures in the Murray–Darling Basin frequently exceed 30 °C even in the absence of drought (B.C. Chessman, personal observations), the warmer conditions during the later stages of the Millennium Drought are likely to have reduced the distribution and abundance of thermally sensitive species, while the cooler conditions after the drought could have allowed some such species to rebound.

Implications and recommendations

While only 11 families fell significantly in prevalence during the drought and did not increase significantly immediately afterwards, other families that lacked a significant post-drought rise might have declined in the early stages of the drought before the monitoring began. Some families were not included in the analysis at all because they were too seldom collected or their representatives were not identified to family level. Even those families that showed no significant drought decline might have included declining species whose responses were masked by the overall resistance of the family.

Consequently, it is possible that many species suffered an adverse effect of the Millennium Drought and had not recovered by the end of the SRA sampling. Such species may still be recuperating but are *prima facie* vulnerable to future droughts.

The observation that several traits were associated with how macroinvertebrate families fared during the Millennium Drought suggests that management actions intended to mitigate deleterious ecological effects of future drought should consider multiple mechanisms of effect. These include loss of flowing water required by rheophilous species, inadequate duration of wetting for species with a long aquatic phase, and effects of high temperatures and hypoxia on species with low physiological tolerances for such conditions. The associations of individual traits with changes in prevalence may have been confounded to some degree by intercorrelation among traits, although such correlations were not especially strong. The notable intercorrelations among atmospheric respiration, DO requirement, rheophily and thermophily are likely to reflect co-evolution of traits in adaptation to different environments (Verberk, Siepel & Esselink, 2008), with thermally intolerant species that require flowing water and high DO concentrations having compounding vulnerabilities to the multiple stressors associated with drought. Controlled experimental studies can help to tease out how these stressors may interact to determine persistence. For example, Verberk & Calosi (2012) have shown that oxygen availability affects upper thermal tolerance in damselfly nymphs.

Trait information can help to identify the species that are most vulnerable to drought and provide insight into the mechanisms through which they are susceptible. However, the availability of such information is limited in most areas. Data sets of freshwater macroinvertebrate traits are reasonably well developed for Europe and North America (Van den Brink *et al.*, 2011), but are grossly deficient for many other regions where climatic drying is forecast. Life-history studies of aquatic invertebrate species, which have fallen out of vogue, need to be revived to fill this void, and dispersal abilities need to be quantified (Robson, Chester & Austin, 2011; Poff, Olden & Strayer, 2012). More physiological information is also needed, such as tolerances of high temperatures and low dissolved oxygen concentrations, and capacity to survive drying. In Australia, such information is beginning to accrue (e.g. Wickson, Chester & Robson, 2012; Chester, Robson & Chambers, 2013; Stewart *et al.*, 2013), but as yet is confined to rather few species.

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